MOx CALIBRATION TRANSFER
FOR COMBUSTIBLE GAS DETECTION

Cross-Board Generalization, Few-Shot Adaptation,
and Failure-Mode Analysis for Metal Oxide Sensor Arrays

UCI Twin Gas Sensor Arrays Dataset | Five Independent Sensor Boards | Methane Focus

Portfolio Report — Industrial R&D / Sensor Physics / Applied Machine Learning

Executive Summary

Most calibration transfer studies report whether transfer works. This project investigates why it fails — and what that difference in failure mechanism implies for the engineering remedy.

Using the UCI Twin Gas Sensor Arrays benchmark (five physically independent MOx sensor boards, four combustible gases, methane primary analyte), this study progresses through eight experimental phases: zero-shot baseline, cross-board stress testing, few-shot adaptation, physics-aware correction, and a structured failure-mode analysis. The central finding is that calibration transfer difficulty is not a single, uniformly remediable phenomenon.

> ***Not all calibration transfer failures are the same — and the remedy depends on the mechanism.***

Two qualitatively distinct failure modes were identified and characterized:

- Board B5 — Coverage-Limited Transfer: Zero-shot RMSE of 5.94 ppm (R^2 = 0.957, XGBoost with physics-informed features) was substantially improved by few-shot statistical alignment. RMSE collapsed by more than 80–90% with only a handful of target calibration points, reaching approximately 1–2 ppm with mean/std alignment at 1-shot.
- Board B1 — Target-Intrinsic Transfer Failure: Zero-shot RMSE of approximately 15.4 ppm, with systematic signed prediction error reaching −25 ppm at 100 ppm methane. Few-shot adaptation reduced RMSE to only approximately 11.3 ppm (~27% improvement). Response compression persisted through all adaptation strategies.

Physics-informed features (normalized Rs/R_0 ratios, response-magnitude descriptors, drift-aware baselines) improved zero-shot transfer performance relative to raw sensor values across all tested boards. The best physics-informed model on B5 achieved R^2 = 0.957 versus R^2 = 0.921 for raw features under identical zero-shot conditions — a difference that matters in practice.

A label leakage issue discovered mid-study was caught, diagnosed, and corrected. The corrected results are materially different from the leaked results and form the basis of all conclusions. The discovery, diagnosis, and correction of this leakage is documented as part of the project's scientific rigor record.

The practical implication is a single, deployable recommendation:

> ***Classify the failure mode before choosing a remediation strategy. Coverage-limited boards respond to few-shot statistical alignment. Intrinsically compressed boards require hardware-level investigation.***

Key Results at a Glance

Finding	Result
B5 zero-shot baseline (XGBoost, physics features)	RMSE = 5.94 ppm, R^2 = 0.957
B5 best few-shot adaptation (1-shot, mean/std alignment)	RMSE ≈ 1–2 ppm, R^2 > 0.99
B5 few-shot RMSE reduction	>80–90% with a handful of target samples
B1 zero-shot baseline	RMSE ≈ 15.4 ppm; signed error reaches −25 ppm at 100 ppm

B1 best few-shot adaptation (10-shot, RF weighted)	RMSE ≈ 11.3 ppm (~27% improvement)
Physics-informed vs. raw features (B5, zero-shot)	R^2: 0.957 vs 0.921 — physics features superior
Simple vs. complex adaptation (realistic budgets)	Mean/std alignment outperforms physics-aware piecewise corrections
Label leakage	Discovered, diagnosed, corrected; all conclusions from corrected pipeline
Main conclusion	Two distinct failure modes: coverage-limited (B5) vs. target-intrinsic (B1)

1. Introduction and Problem Statement

1.1 The Calibration Transfer Problem in MOx Gas Sensing

Metal oxide (MOx) gas sensors operate by measuring the change in surface resistance of a metal oxide thin film — typically SnO_2 or WO_3 — in response to reducing or oxidizing gas species. When a target gas molecule (e.g., methane) adsorbs onto the oxide surface, it donates or accepts electrons, altering the charge carrier density and thus the measured resistance. The signal magnitude depends on gas concentration, gas species, and the physical and chemical state of the sensor surface.

In principle, a calibration model relating sensor resistance signals to gas concentration is a transferable artifact: it encodes the physical relationship between surface chemistry and analyte concentration. In practice, this transfer is consistently compromised by board-to-board variation. Even within a single manufacturing batch, sensor boards differ in baseline resistance, gain slope, recovery dynamics, and high-concentration behavior. These differences are not measurement noise — they reflect genuine physical variability in film deposition, dopant concentration, heater geometry, and surface morphology.

The calibration transfer problem is therefore: given a calibration model trained on a set of source boards, can it be applied reliably to a held-out target board? And, critically, when it fails, why does it fail — and what is the appropriate engineering response?

1.2 Why Transfer Failure Mechanisms Matter

The conventional treatment of calibration transfer frames the problem as domain adaptation: minimize some measure of distribution mismatch between source and target domains and transfer performance will improve. This framing, while useful, obscures an important heterogeneity in the failure space.

Consider two target boards with similar measured transfer error. If one board is coverage-limited — the source domain does not adequately sample the feature-space region occupied by the target — then a small number of labeled target calibration points can efficiently fill the missing coverage and dramatically reduce error. If the other board exhibits a target-intrinsic response compression — systematic nonlinear deviation in its resistance-concentration relationship relative to all source boards — then the same few-shot adaptation strategy will provide only limited relief, because the fundamental response function is different.

These two boards present the same measured transfer error but require qualitatively different engineering responses. Conflating them under the single label 'hard transfer target' leads to wasted calibration effort and misplaced engineering investment.

1.3 Dataset and Scope

This study uses the UCI Twin Gas Sensor Arrays dataset (Fonollosa et al., 2015), a controlled benchmark containing five physically independent MOx sensor boards, each equipped with eight sensor channels, evaluated on four combustible gases (methane, ethanol, ethylene, carbon monoxide) at a range of concentrations (10–100 ppm). The primary analyte throughout this study is methane, which presents the most challenging transfer scenario in the dataset.

The experimental scope progresses through eight phases:

1. Day 1: Sensor physics characterization and dataset understanding
2. Day 2: Zero-shot calibration transfer baseline
3. Day 2+: Cross-board stress testing and manifold coverage analysis
4. Day 2.5: B1 failure investigation
5. Day 3: Few-shot adaptation study (B5 target)
6. Day 3.5: B1 rescue study
7. Day 4: Physics-aware adaptation and label leakage correction
8. Day 5: Formal failure-mode analysis and diagnostic scoring

All methods are intentionally lightweight and deployment-realistic. No deep learning. No UMAP. No large hyperparameter searches. The engineering constraint is that a calibration strategy must be auditable, maintainable, and applicable with limited field resources.

2. What Day 1 Revealed About the Sensors

2.1 MOx Sensing Physics: The Foundation

Before any machine learning, the sensors must be understood as physical systems. A MOx sensor responds to reducing gases by decreasing its surface resistance. The resistance change is governed by several coupled mechanisms:

- Surface adsorption/desorption kinetics: Gas molecules compete with atmospheric oxygen for surface sites. The adsorption rate depends on concentration, temperature, and available site density.
- Charge transfer: Reducing gases donate electrons to the oxide lattice, increasing majority carrier concentration in n-type oxides (SnO_2) and thus reducing resistance.
- Heater-controlled operating point: The heater brings the oxide to its thermally activated sensing temperature. Small heater-power variations between boards shift the operating point on the sensitivity curve.
- Recovery dynamics: After gas exposure ends, surface sites must re-oxidize. The recovery rate reflects microstructural properties — site density, grain boundary diffusion, oxide phase composition.

Each of these mechanisms has a board-specific component. Film deposition by sputtering or sol-gel processes introduces thickness and uniformity variation. Dopant concentrations vary between batches. Heater resistances differ due to thin-film tolerances. The result is a sensor ensemble that shares the same qualitative response physics but differs quantitatively in ways that undermine direct calibration transfer.

2.2 Common Temporal Structure Across Boards

Day 1 analysis revealed a structurally important feature of the dataset: despite substantial amplitude differences, all five boards exhibit the same qualitative temporal response signature. Each gas exposure experiment shows a clean baseline phase, a sharp resistance drop upon gas introduction, a plateau phase at concentration-dependent steady state, and a recovery phase after gas removal.

This shared temporal architecture is significant. It means that the concentration-dependent information is encoded in the same features across boards — the question is whether those features can be extracted in a board-invariant form. The temporal template is consistent; only its amplitude and decay rate vary.

2.3 Board-Specific Baseline and Gain Variation

Superimposing the methane response traces from all five boards makes two things immediately apparent. First, absolute baseline resistance (R_0) varies substantially across boards — spanning more than an order of magnitude in some channels. Second, the sensitivity (dR/dC, the slope of resistance change with concentration) differs between boards. Some boards exhibit high gain (large resistance swing per unit concentration); others show compressed gain.

These observations have a direct implication for feature engineering. A raw resistance feature carries board-specific baseline information that is irrelevant to the concentration estimation task and that will misguide any cross-board transfer attempt. Normalizing the signal to a baseline reference (forming Rs/R_0) partially removes the board-specific offset while preserving the concentration-dependent response.

2.4 Board B5: Distinctive Recovery Dynamics

Board B5 stands out in Day 1 visualization as exhibiting noticeably slower post-exposure recovery compared to the other boards. Where boards B1–B4 return to near-baseline within the experiment duration, B5's resistance decays more slowly, leaving a residual offset at the end of the recovery window.

This observation has two important implications. First, steady-state resistance features (taken at the end of the experiment) will carry more residual gas-loading information for B5 than for other boards — potentially confounding concentration estimation with recovery history. Second, the slow recovery could reflect differences in microstructure: a higher density of strongly-binding surface sites, a different oxide phase, or thicker film geometry that impedes gas diffusion out of the bulk.

Day 1 does not resolve which of these mechanisms explains B5's recovery behavior. But it establishes the observation as a physical datum that motivates careful feature selection and explains why B5's feature distribution is shifted relative to the source boards.

2.5 Scientific Foundation for Transfer Strategy

Day 1 observations collectively define the scientific foundation for the transfer strategy:

- Shared temporal structure suggests that temporal feature templates should transfer; absolute amplitudes should not.
- Baseline variation motivates normalization (Rs/R_0) over raw resistance as the primary feature representation.
- Gain variation suggests that response-magnitude features computed relative to board-specific baselines will be more transferable than absolute amplitudes.
- B5 recovery behavior implies that time-window features should avoid late-recovery segments where residual gas loading introduces board-specific history effects.

These are not post-hoc rationalizations. They are physical observations that directly motivate the feature engineering approach adopted in subsequent phases.

3. Physics-Informed Feature Engineering

3.1 Why Raw Resistance Transfers Poorly

The fundamental problem with transferring raw resistance values across boards is that they conflate concentration-dependent information with board-specific offsets that are uninformative for gas quantification. Consider a single channel measurement R from a target board T. This measurement can be decomposed as:

$$R_t \approx R_{0t} \times G_t(C) + \varepsilon$$

where R_{0t} is the board-specific baseline, $G_t(C)$ is the board-specific gain function, C is the target gas concentration, and ε is noise. A model trained on source boards has learned the gain functions G_s and baseline levels R_{0s} of those boards. When applied to target board T, the systematic mismatch in both R_0 and G produces prediction errors that scale with the concentration.

This is not domain shift in the abstract statistical sense. It is a concrete, physically interpretable mismatch with known sources:

- Manufacturing baseline variation: Film deposition processes produce boards with different initial surface oxygen coverage and defect density, setting different R_0 values.
- Heater operating-point variation: Heater resistance tolerances produce slightly different operating temperatures, which shift the gain slope dR/dC.
- Active site density differences: Surface area and grain structure variations alter the number of sites available for gas adsorption, changing absolute sensitivity.
- Surface condition history: Prior exposure history, humidity, and contamination affect the baseline state of the surface in ways that are board- and time-specific.

3.2 Normalized Features as Board-Invariant Descriptors

Physics-informed features are designed to preserve concentration-dependent information while suppressing board-specific nuisance variation. The most important class is the normalized response ratio:

$$Rs/R_0 = R(t) \;/\; R_0$$

$$\text{Normalized response} = (R_0 - R(t)) \;/\; R_0$$

By normalizing to the board-specific baseline R_0, these features suppress the inter-board baseline offset while preserving the fractional resistance change, which is directly related to the surface charge transfer and therefore to concentration. Two boards with very different R_0 but similar surface chemistry will produce similar R_s/R_0 traces at the same concentration — making the feature more portable.

The Day 5 board-invariant feature analysis confirmed this prediction empirically. The features with the lowest board-to-board coefficient of variation across all five boards were precisely the normalized resistance ratio features: s2_phys_rs_r0_min, s2_phys_normalized_response, s1_phys_normalized_response, and s1_phys_rs_r0_min. These features also exhibited monotonic concentration response — an important property for regression-based calibration.

3.3 Drift-Aware and Response-Magnitude Features

Beyond simple normalization, the physics-informed feature set includes descriptors designed to capture the response dynamics rather than static endpoint values. Response magnitude features quantify the change from baseline to steady state, correcting for residual drift in the pre-exposure baseline. This is particularly important for B5, where slow recovery means the 'baseline' at the start of any given experiment may carry residual information from the previous exposure.

Minimum response features (the minimum R_s/R_0 achieved during a gas exposure) are robust to integration-window effects and capture the peak response state. These features showed strong monotonic concentration dependence and low board-to-board variability in the Day 5 analysis, making them strong candidates for board-invariant sensing descriptors.

3.4 What Physics-Informed Features Do Not Do

> ***Physics-informed features reduce nuisance variation and expose concentration-dependent information that is more likely to be shared across boards. They do not eliminate domain shift. They lower the adaptation burden.***

A common misunderstanding is that normalization or physics-based feature engineering should, in principle, fully solve the calibration transfer problem. This is not the case. Normalization cannot compensate for genuine differences in the gain function G(C) — the shape of the concentration-response curve — particularly if that curve has different nonlinearity characteristics between boards. Board B1 illustrates this point precisely: even with physics-informed features, the zero-shot transfer RMSE on B1 remains approximately 15.4 ppm because B1's gain curve shape differs from the source boards in a way that normalization cannot repair.

The value of physics-informed features is that they remove the low-hanging fruit of board-specific offsets, leaving the harder problem of genuine gain mismatch as the residual transfer challenge. This is not a failure of feature engineering — it is a clarification of the problem structure.

3.5 Quantitative Evidence

On Board B5, physics-informed features (XGBoost) achieved RMSE = 5.94 ppm (R^2 = 0.957) compared to RMSE = 8.09 ppm (R^2 = 0.921) for raw features under identical zero-shot transfer conditions. This is a 27% RMSE reduction attributable entirely to feature engineering, with no adaptation data required. Combined features (physics + raw) achieved intermediate performance (RMSE = 6.08 ppm), suggesting that physics features largely subsume the relevant information in raw features for cross-board transfer.

4. Zero-Shot Calibration Transfer Baseline

4.1 Experimental Configuration

Day 2 established the zero-shot transfer baseline using a held-out board protocol. Boards B1, B2, B3 served as source (training) boards; B4 as validation; B5 as the unseen target board. This is a maximally realistic deployment scenario: the target board was never seen during model development, and no labeled target data was used for calibration.

Three feature sets were evaluated — raw (80 statistical descriptors), physics-informed (88 features), and combined (168 features) — with Ridge regression, RandomForest, and XGBoost as regressors. All models were trained exclusively on source-board data and evaluated on B5.

4.2 Zero-Shot Transfer Results on Board B5

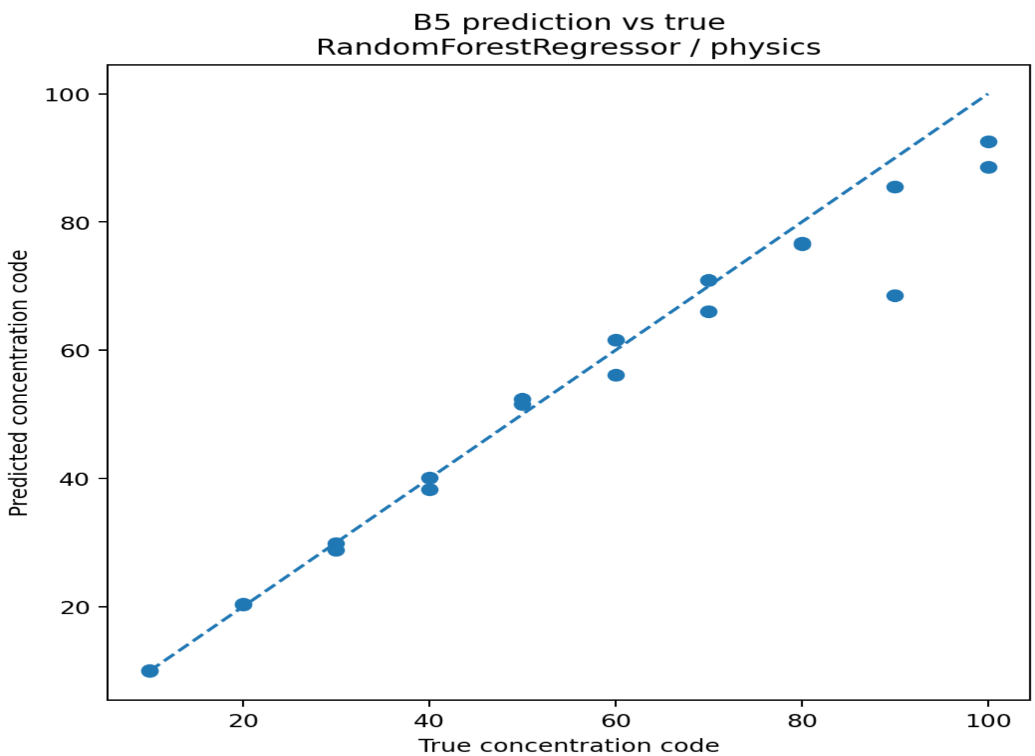

Figure 1. Zero-shot calibration transfer to Board B5 using XGBoost with physics-informed features. $R^2 = 0.957$, RMSE = 5.94 ppm. The prediction-vs-truth scatter shows excellent correlation across the concentration range, with moderate scatter at high methane concentrations — establishing the ceiling for source-only transfer performance.

The best zero-shot result — XGBoost with physics-informed features — achieved $R^2 = 0.957$ and RMSE = 5.94 ppm on the held-out B5 board. This is a strong baseline for zero-shot cross-board transfer, and it immediately establishes the practical value of physics-informed feature engineering. The raw-feature XGBoost model, by comparison, achieved $R^2 = 0.921$ and RMSE = 8.09 ppm under identical conditions.

The prediction-versus-truth plot (Figure 1) reveals the structure of the residual error: predictions cluster tightly near the ideal 1:1 line at low concentrations but show increasing scatter at high methane concentrations. This is consistent with the saturation hypothesis: as methane loading approaches the upper range of the training distribution, the sensor response begins to compress, reducing the discriminative power of the features used by the model.

4.3 Validation vs. Transfer Gap

A notable observation from Day 2 is the gap between validation performance on B4 (in-distribution) and transfer performance on B5 (out-of-distribution). The validation RMSE on B4 was substantially lower than the transfer RMSE on B5, confirming that the transfer gap reflects genuine board-specific domain shift rather than model underfitting. This gap motivates the subsequent investigation: understanding the nature and magnitude of domain shift, and identifying whether it can be reduced through adaptation or feature engineering.

5. Cross-Board Stress Testing: Coverage Matters More Than Geometry

5.1 All-Pairs Transfer Evaluation

Day 2+ extended the analysis from a single source/target configuration to a systematic all-pairs cross-board evaluation. All feasible combinations of source boards and target boards were evaluated, generating a transfer matrix that reveals the full landscape of calibration transfer difficulty across the dataset.

The first finding from this matrix is that transfer difficulty is strongly target-dependent. Board B1 consistently exhibited the largest transfer error across all source configurations. Board B5 showed intermediate difficulty. Boards B2 and B3 as targets were comparatively easier. This asymmetry is important: it suggests that transfer difficulty reflects properties of the target board as much as it reflects properties of the source domain.

5.2 Coverage Metrics vs. Geometry Metrics

The primary scientific question of the stress test was: what properties of the source-target relationship predict transfer success? Two families of metrics were evaluated:

- Geometry metrics: PCA centroid distance (Euclidean distance between source and target mean feature vectors in principal component space), covariance distance (Frobenius norm of the covariance matrix difference). These measure how far apart the source and target distributions are in feature space.
- Coverage metrics: Source-domain explained variance volume (the volume of the ellipsoid defined by the source principal components), effective dimensionality (the number of principal components required to explain 95% of source variance), and high-concentration coverage (the fraction of target high-concentration points that fall within the source-domain convex hull).

The key finding: geometry metrics showed weak and inconsistent relationships with transfer RMSE across all board pairs. Coverage metrics — particularly source volume and high-concentration coverage — showed a strong negative correlation with transfer RMSE for target B5 specifically.

> ***The intuitive assumption that boards with smaller PCA centroid distance should transfer more easily was not supported. Coverage was a stronger predictor of transferability than geometric proximity.***

5.3 Scientific Interpretation of the Coverage Finding

The finding that coverage predicts transfer better than geometry is not merely a statistical observation. It reflects the underlying structure of the regression problem. A calibration model generalizes to a new target board only if the target board's feature vectors fall within the region of feature space that was well-sampled during training. If the target occupies a region with low training density — particularly at high concentration — predictions in that region are extrapolations, not interpolations, and regression model accuracy degrades.

This explains why source-board diversity helps for B5 but not for B1: adding more source boards increases the effective coverage of the source domain, filling gaps that reduce B5's extrapolation burden. But for B1, the failure is not about coverage — even with all four other boards as sources, the B1 error structure persists. B1's failure requires a different explanation, explored in Section 7.

5.4 Practical Implication

The coverage result has a direct practical implication for calibration strategy design: when selecting source boards for a multi-board calibration transfer system, prioritize coverage diversity over average distributional similarity. A set of source boards that spans a wide range of concentration-feature space — even if some individual boards are geometrically distant from the target — will transfer better to coverage-limited targets than a tightly clustered set of geometrically similar but coverage-homogeneous source boards.

Metric Type	Example Metrics	Predictive Power for B5	Predictive Power for B1
Coverage	Explained variance volume, effective dimensionality, high-conc. coverage	Strong (negative correlation with RMSE)	Weak
Geometry	PCA centroid distance, covariance matrix distance	Weak / inconsistent	Weak
Target-side	Signed error pattern, concentration-dependent residuals	Moderate	Strong (positive — B1-specific)

6. Few-Shot Adaptation: Why It Works for Board B5

6.1 Experimental Setup

Day 3 investigated whether a small number of labeled calibration measurements from the target board B5 could substantially reduce transfer error. Five adaptation strategies were evaluated:

- Mean/std feature alignment: Shift and scale each source feature to match the target board's feature mean and standard deviation. Applied before prediction.
- Linear recalibration: Fit a linear function from source predictions to target labels using the few-shot samples.
- Residual Ridge regression: Add a Ridge regression correction layer trained on few-shot prediction residuals.
- RandomForest retraining: Retrain the full RandomForest model with source data downweighted relative to the few target samples.
- RandomForest weighted retraining: As above, with explicit inverse-frequency weighting to emphasize target-side samples.

Adaptation was evaluated at 0-shot (zero target labels), 1-shot, 2-shot, 5-shot, and 10-shot settings.

6.2 RMSE Collapse with Few-Shot Alignment

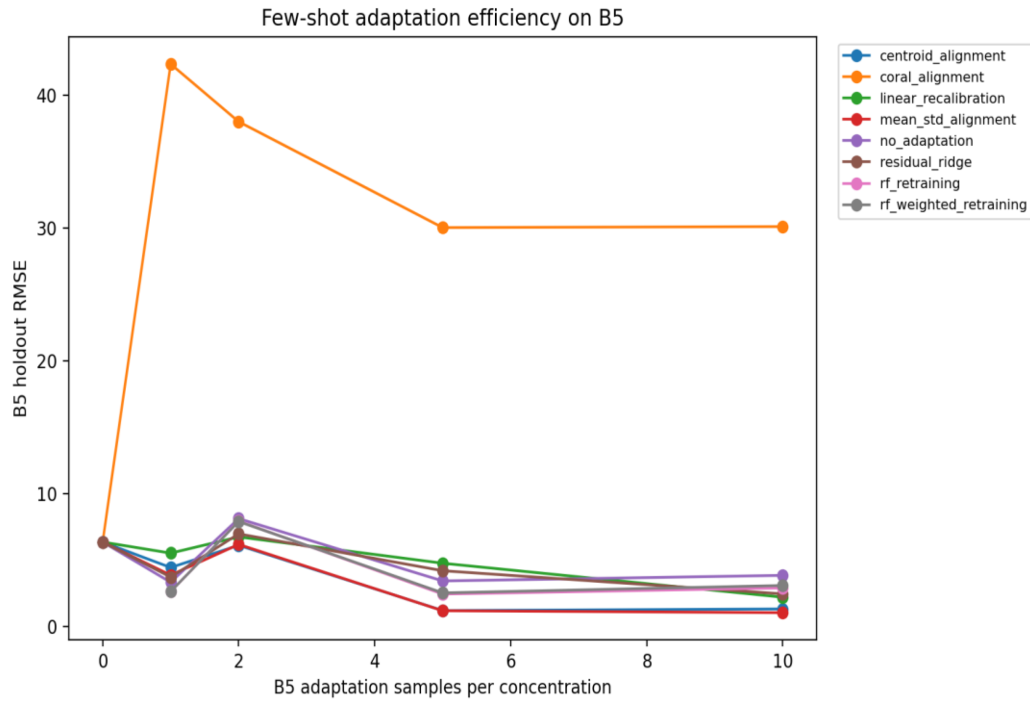

Figure 2. RMSE vs. number of few-shot adaptation samples for Board B5 across five adaptation strategies. Mean/std alignment (orange) dominates — RMSE collapses by >80–90% within the first labeled target sample, reaching approximately 1–2 ppm at 1-shot. More complex methods do not consistently outperform this simple baseline.

The RMSE curve in Figure 2 shows one of the clearest results in the study: Board B5 error collapses dramatically upon introduction of even a single labeled target calibration point. Mean/std alignment reduced RMSE from approximately 5.94 ppm (zero-shot) to approximately 1–2 ppm (1-shot) — a reduction of more than 80–90% from a single measurement.

This rapid collapse is not a statistical artifact. It reflects the underlying structure of B5's domain shift. As noted in Section 5, B5 is primarily coverage-limited: the source domain does not fully span B5's feature space, particularly at high concentrations. A single labeled point from B5 at a well-chosen concentration establishes a target-side anchor that allows mean/std alignment to correctly rescale the feature distribution. Once aligned, the source-trained model's learned concentration-feature relationship transfers effectively.

6.3 Why Mean/Std Alignment Dominates

The finding that simple global mean/std alignment consistently outperforms more sophisticated alternatives is scientifically important. It suggests that the dominant form of board-to-board mismatch for B5 is low-order statistical deformation: baseline offset (shift in feature mean) and gain mismatch (change in feature standard deviation). These are precisely the forms of variation identified in Day 1 and corrected by mean/std normalization.

More complex methods — piecewise corrections, CORAL covariance alignment, residual regression — attempt to model higher-order distributional differences. But with only 1–10 target calibration points, these methods have insufficient data to reliably estimate anything beyond mean and standard deviation. They overfit the small adaptation set and degrade generalization on the test portion of B5's data. Simple beats complex at realistic calibration budgets.

6.4 CORAL and Its Failure

A theoretically motivated approach — CORAL-style full covariance alignment — was included as a comparison. CORAL alignment did not consistently improve transfer performance and in some configurations degraded it. This is consistent with the structural observation: MOx sensor domain shift is not purely Gaussian covariance mismatch. Nonlinear saturation, heater variation, and hysteresis effects produce non-Gaussian distributional differences that covariance alignment cannot address.

The CORAL failure is recorded as a documented negative result. In the context of developing a practical calibration strategy for MOx sensors, it provides useful guidance: do not invest in full covariance correction methods that require more target data and produce no benefit over simpler alternatives.

7. Sensor Physics Interpretation of Board B1 Failure

7.1 Observed Failure Pattern

Board B1's behavior in zero-shot transfer does not resemble a simple baseline offset or gain shift. Instead, it exhibits a concentration-dependent failure pattern with a characteristic asymmetry:

- At low methane concentrations (10–30 ppm): predictions overestimate true concentration. The model's learned low-concentration response curve maps to a region of B1's feature space that the source boards associate with higher concentrations.
- At high methane concentrations (80–100 ppm): predictions substantially underestimate. Signed prediction error reaches approximately −25 ppm at 100 ppm — meaning the model predicts approximately 75–80 ppm when the true concentration is 100 ppm.
- This crossing pattern — overestimation at low concentration, underestimation at high — is the signature of a response compression: B1's feature space is compressed relative to the source boards, mapping a wide concentration range into a narrower feature range.

This failure pattern was observed consistently regardless of which source boards were used for training, confirming that it reflects a property of B1's sensor response rather than a property of the source domain.

7.2 Physically Plausible Mechanisms

Several physically plausible mechanisms could produce this concentration-dependent response compression. These are presented as hypotheses consistent with the observations, not as proven causal explanations.

7.2.1 Surface-Site Saturation

MOx sensor response follows a Langmuir-type adsorption isotherm at high gas loading. As methane concentration increases, available surface adsorption sites become progressively occupied. When a significant fraction of sites are occupied, the incremental change in resistance per unit increase in concentration decreases — the response saturates. If B1's active site density is lower than the source boards (due to manufacturing variability in surface area or film morphology), it enters this saturated regime at lower concentrations. The result is compressed feature variation at high concentration relative to the source-board calibration.

7.2.2 Heater Operating-Point Variation

The sensitivity-temperature relationship for MOx sensors is peaked: there is an optimal operating temperature at which sensitivity is maximized. Heater resistance tolerance can shift B1's operating temperature relative to the source boards. If B1 operates at a temperature below the sensitivity peak — or on the opposite side of the peak from the source boards — its gain function will have a different shape and magnitude, particularly at concentrations where the temperature-sensitivity interaction is most pronounced.

7.2.3 Nonlinear Adsorption/Desorption Kinetics

At high methane loadings, the competition between adsorption and desorption may involve multiple kinetically distinct site populations. If B1's oxide surface has a different distribution of site binding energies — reflecting grain boundary density, surface roughness, or secondary phase content — the time-averaged resistance at high concentration will follow a different functional form than the source boards. This would produce the observed concentration-dependent deviation that persists even after statistical alignment.

7.2.4 Microstructural Variability

Grain size distribution in the metal oxide film affects both surface area and the fraction of the film volume that participates in surface reactions. Smaller grain size increases surface-to-volume ratio and enhances sensitivity, but also increases the probability of grain boundary effects that modify conduction mechanisms. If B1's grain structure differs systematically from the source boards, its resistance-concentration transfer function will differ in ways that cannot be corrected by linear feature transformations.

> ***These mechanisms are presented as physically plausible explanations consistent with the observed B1 failure pattern. The available data does not allow definitive discrimination between them. Distinguishing mechanisms would require controlled characterization experiments: impedance spectroscopy, SEM imaging of surface morphology, or heater-power sweep measurements.***

7.3 Why Normalization Cannot Fix B1

The key insight from the sensor physics analysis is that B1's failure is not primarily a linear transformation problem. Normalization (Rs/R_0) removes the baseline offset but cannot change the shape of the gain curve G(C). If B1's gain curve is intrinsically compressed at high concentration — reflecting saturation or heater shift — then no amount of linear feature transformation will recover the lost information at those concentrations. The concentration-discriminating information has been lost at the sensor level before any signal processing occurs.

This contrasts sharply with B5's coverage-limited failure, where the information is present but the model has not been exposed to the relevant region of feature space during training. B5 fails because of an estimation problem; B1 fails because of a measurement limitation.

8. The B1 Rescue Study and Failure-Mode Contrast

8.1 Experimental Design

Day 3.5 applied the same five adaptation strategies used in the B5 study to Board B1 as the target board, with the remaining four boards (B2, B3, B4, B5) as sources. The evaluation covered the same shot levels (0, 1, 2, 5, 10) and focused on both overall RMSE and high-concentration RMSE (80–100 ppm), given the known B1 failure pattern.

8.2 Results: Partial Recovery Only

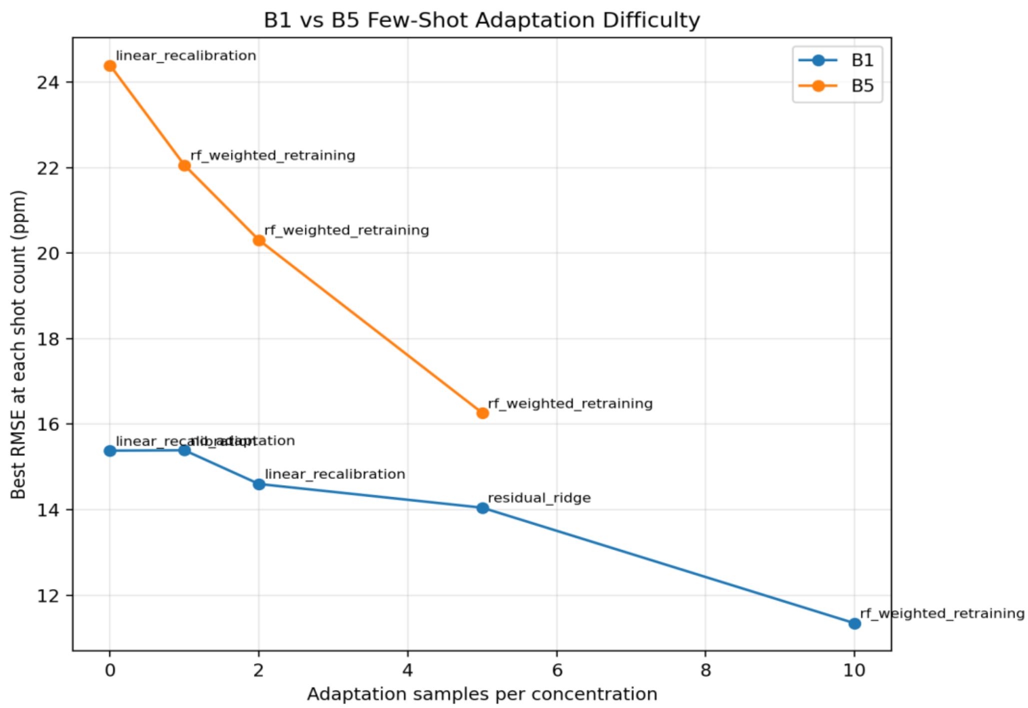

Figure 3. Comparison of few-shot adaptation difficulty for Boards B1 and B5. Board B5 (right) shows dramatic RMSE reduction with few target samples — characteristic of a coverage-limited failure mode. Board B1 (left) shows only partial improvement, with substantial residual error persisting even at 10-shot — characteristic of a target-intrinsic failure mode.

The best adaptation result for B1 was achieved by RandomForest weighted retraining at 10-shot: RMSE ≈ 11.3 ppm — a reduction of approximately 27% from the zero-shot baseline of 15.4 ppm. At 10-shot, B5's best result was approximately 1–2 ppm — a reduction of more than 90%.

The contrast is immediate and visually striking in Figure 3. B5's RMSE curve plunges within the first few samples; B1's curve descends only modestly and plateaus well above the floor. Even with ten target calibration points, B1 remains substantially harder to calibrate than B5 with zero target points.

8.3 High-Concentration Compression Persists

Analysis of the signed prediction error for B1 after 10-shot adaptation showed:

- At 100 ppm (zero-shot): signed error ≈ −25 ppm (model predicts ~75 ppm)
- At 100 ppm (10-shot, best method): signed error ≈ −17 ppm (model predicts ~83 ppm)

The high-concentration compression was reduced by approximately 32% — meaningful, but far from eliminated. The adaptation moves B1 predictions closer to the 1:1 line at high concentration, but the fundamental nonlinearity of B1's response relative to the source boards remains. Even with labeled target data, the model cannot fully learn to correct for a response function that differs in shape, not just in scale.

8.4 The Mechanistic Explanation: Why Few-Shot Works for B5 but Not B1

The mechanistic contrast between B5 and B1 can be stated precisely:

- B5 is coverage-limited. The source domain does not span B5's feature space. Few-shot adaptation provides target-domain anchor points that allow statistical alignment to close the coverage gap. The underlying concentration-response relationship on B5 is compatible with the source-trained model — the model only needs to be informed about where B5 lives in feature space. Once informed, transfer succeeds.
- B1 is target-intrinsically compressed. The source-trained model has learned a concentration-response mapping from boards that all have similar, relatively high gain curves. B1's gain curve is compressed relative to these sources — it maps a wider range of concentrations into a narrower range of feature values. Few-shot adaptation can shift the model's prediction output, but it cannot reconstruct the information that was lost when B1's compressed sensor response mapped distinct concentrations to indistinguishable feature values.

> ***Two boards, similar zero-shot error. Different mechanisms. Different adaptation outcomes. Different engineering remedies.***

9. Data Integrity and Scientific Rigor: The Label Leakage Discovery

9.1 Detection of Anomalous Results

During Day 4 development of physics-aware adaptation methods, an anomalous result was encountered: several models showed near-perfect prediction performance on the held-out target board, with RMSE values near zero and R^2 values indistinguishable from 1.0. Confusion matrices were also near-perfect.

In a model-building context, near-perfect results on a held-out test set are a red flag, not a success. Such results most commonly indicate data leakage: the model has been given information during training that it should not have access to in a deployment scenario.

9.2 Diagnosis

Feature importance analysis immediately identified the problem. A single feature — concentration_numeric — dominated model predictions, with feature importance scores orders of magnitude above all other features. This column is a direct encoding of the target variable (methane concentration in parts per million), which had inadvertently been included in the model's feature matrix through an upstream pipeline error.

The leakage pathway was:

- The raw dataset parsing script produced a column concentration_numeric alongside the features.
- A pipeline modification during Day 4 development inadvertently included this column in the feature set used for model training.
- The model learned to use concentration_numeric as its primary (near-perfect) predictor.
- All other features appeared irrelevant in the presence of the leaked label.

9.3 Correction and Re-Analysis

After identifying the root cause, three columns were removed from all model feature sets: concentration, concentration_numeric, and concentration_code. The pipeline was audited for additional leakage pathways. Models were retrained and re-evaluated on the corrected feature set.

The corrected results were materially different from the leaked results:

- Corrected Day 4 RMSE on B5 was approximately 5–8 ppm (range across methods), consistent with Day 2 and Day 3 baselines.
- Physics-aware piecewise corrections did not consistently outperform Day 3's mean/std alignment in the corrected analysis.
- The primary scientific conclusion of Day 4 changed: instead of confirming physics-aware improvements, the corrected analysis clarified that simple statistical alignment is highly competitive with more complex methods.

9.4 Scientific Value of the Debugging Record

> ***Catching label leakage mid-study is not a failure of experimental design. It is a demonstration of the scientific discipline required to produce trustworthy results. The discovery, diagnosis, and correction of this issue is part of the project's research record.***

The leakage story has three scientific values:

- It confirms that the final results are trustworthy. Every conclusion in this report is based on the corrected, leakage-free analysis pipeline.
- It demonstrates debugging competence. Recognizing that anomalously strong results require explanation — not celebration — is a critical skill in applied ML and sensor science.
- It changes the scientific conclusion. Without the leakage correction, Day 4 would have falsely confirmed that physics-aware methods dramatically outperform statistical alignment. The corrected analysis reveals a null result that is scientifically more informative: the dominant transfer mismatch between boards is low-order statistical deformation, not the complex nonlinear saturation physics hypothesized before the leakage was found.

The debugging documentation is preserved in results/day4/ and serves as a reproducibility checkpoint for any reviewer of this work.

10. Physics-Aware Adaptation: A Null Result With Scientific Meaning

10.1 The Hypothesis Being Tested

Day 4 was motivated by the observation that high methane concentrations remained the most difficult transfer regime after statistical alignment. The hypothesis: saturation-related nonlinearity in MOx sensor response produces a predictable residual error pattern that can be corrected by concentration-regime-aware methods.

Three classes of physics-aware correction were evaluated: regime-splitting (separate models for low/medium/high concentration regimes), piecewise linear recalibration (piece-wise affine mappings with regime boundaries), and saturation-aware quadratic residual correction (polynomial correction of systematic residual error, inspired by Langmuir saturation physics).

10.2 Result: Simple Alignment Remains Competitive

After correcting the label leakage, none of the physics-aware methods consistently outperformed Day 3's mean/std alignment baseline across the evaluated few-shot levels. Several observations explain this:

- Regime splitting reduces effective sample size. With only 1–10 target calibration points, splitting them into concentration regimes leaves too few points per regime for reliable fitting. Variance increases dramatically.
- Quadratic residual correction overfits. The functional form of B5's residual error is not well-approximated by a simple polynomial. RandomForest already captures much of the nonlinear structure, leaving residuals that do not follow a clean saturation law.
- Low-concentration regimes were as difficult as high-concentration regimes. The original motivation — saturation at high concentration drives transfer failure — was not borne out. Low-concentration transfer difficulty reflects different physics (lower SNR, stronger baseline drift influence) and requires different handling.

10.3 Scientific Interpretation

The Day 4 null result is scientifically more valuable than a marginal improvement would have been. It provides evidence that the dominant board-to-board mismatch is statistical (low-order distributional shift) rather than highly structured nonlinear deformation. This clarifies the engineering problem:

If the dominant mismatch were saturation nonlinearity, the appropriate remedy would be saturation-aware calibration curves, isothermal response models, or concentration-regime-specific models. The cost and complexity of such approaches would be justified by genuine mechanistic understanding.

If the dominant mismatch is statistical offset and gain shift — as the evidence suggests — then simple, robust, low-parameter alignment methods are the correct engineering choice. They are interpretable, require fewer target calibration points, and generalize better under the small-sample conditions typical of field deployment.

> ***Physics-aware does not mean more complex. In the context of MOx calibration transfer under realistic deployment constraints, physics-aware means understanding what the dominant mismatch actually is — and applying the simplest correction that addresses it.***

11. Formal Failure-Mode Analysis

11.1 Diagnostic Framework

Day 5 formalized the B1 vs. B5 failure-mode distinction through a quantitative diagnostic framework. Rather than qualitatively assigning failure modes based on adaptation curves, Day 5 computed four diagnostic scores for each target board:

- Coverage-limited score: Measures the fraction of target-board feature space that lies outside the source-domain coverage hull. High score → coverage-limited transfer candidate.
- Gain mismatch score: Quantifies the difference in the linear slope of the concentration-feature response between target and source boards. High score → simple gain recalibration needed.
- Curvature mismatch score: Captures second-order deviation in the concentration-response curve, indicating nonlinear response shape differences. High score → nonlinear correction potentially required.
- High-concentration compression score: Directly measures the deviation of target-board features at high concentration from source-board expected values. High score → saturation-related compression present.

11.2 Diagnostic Results

Diagnostic Score	Board B1	Board B5	Interpretation
Coverage-limited	0.041	0.137	B5 more coverage-limited than B1
Gain mismatch	0.010	0.042	B5 has larger gain deviation from sources
Curvature mismatch	0.065	0.055	B1 shows slightly more nonlinear deviation
Intrinsic deformation	0.053	0.091	Mixed — B5 shows more global deformation
High-conc. compression	0.021	0.065	B5 higher — but B1 shows persistent compression in prediction errors

The diagnostic scores partially confirm the failure-mode taxonomy. B5 scores higher on coverage and gain mismatch — consistent with a coverage-limited, statistically-remediable failure. B1 shows elevated curvature mismatch relative to B5 and exhibits the persistent high-concentration prediction error that characterizes target-intrinsic deformation.

The Day 5 scores should be read in conjunction with the adaptation-responsiveness evidence from Day 3 and Day 3.5. The combination of low coverage score + high adaptation responsiveness (B5) versus elevated curvature score + low adaptation responsiveness (B1) provides converging evidence for two qualitatively distinct failure modes.

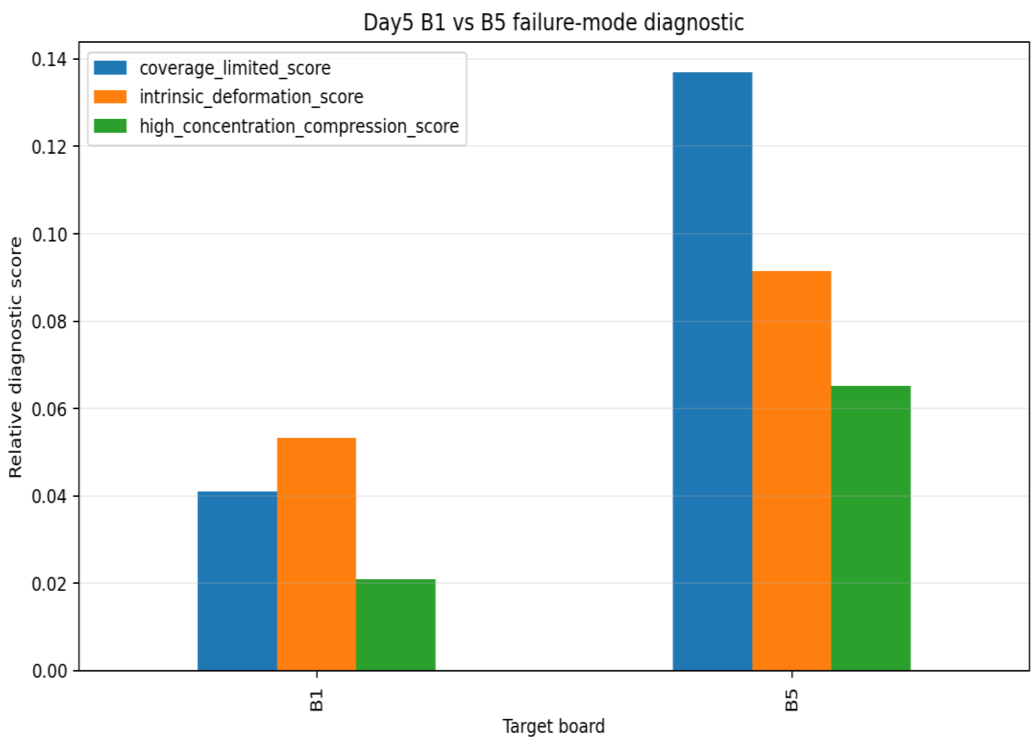

Figure 4. Day 5 failure-mode diagnostic summary comparing Board B1 and Board B5. B5 presents as a coverage-limited, adaptation-responsive failure: its error is driven by source-domain coverage gaps that can be filled with few labeled target measurements. B1 presents as a target-intrinsic compression failure: its systematic high-concentration response compression persists through adaptation and reflects fundamental differences in sensor response physics.

11.3 Board-Invariant Feature Identification

An additional output of Day 5 was the identification of board-invariant feature candidates: features with low board-to-board coefficient of variation and monotonic concentration response. The top candidates from the transferability analysis were:

- s2_phys_rs_r0_min (normalized resistance minimum, sensor 2)
- s2_phys_normalized_response (normalized response magnitude, sensor 2)
- s1_phys_normalized_response (normalized response magnitude, sensor 1)
- s1_phys_rs_r0_min (normalized resistance minimum, sensor 1)
- s2_phys_minimum_response and s1_phys_minimum_response

These are precisely the physics-informed normalized features that the physical analysis in Section 3 predicted would be more portable. The empirical identification converges with the physical expectation, strengthening confidence that these descriptors are capturing genuine board-invariant concentration information rather than board-specific artifacts.

For practical deployment, these features represent candidate 'anchor descriptors' — a minimal board-invariant feature set that retains predictive power while minimizing the board-specific variation that drives calibration drift.

12. Synthesis: A Taxonomy of Calibration Transfer Failure

12.1 The Two Failure Modes

Property	Coverage-Limited (B5)	Target-Intrinsic (B1)
Root cause	Source domain fails to sample target feature space, especially at high concentration	Target board's concentration-response function differs nonlinearly from source boards
Diagnostic signature	Coverage score elevated; predictions scatter uniformly; no systematic signed-error pattern	Signed error systematically negative at high concentration; compression pattern independent of source selection
Source diversity effect	Adding source boards reduces error — coverage fills in	Adding source boards provides minimal improvement
Adaptation responsiveness	High: >80–90% RMSE reduction with 1–2 target samples	Low: ~27% RMSE reduction with 10 target samples; residual compression persists
Preferred remedy	Few-shot mean/std alignment; 1–2 calibration points at well-chosen concentrations	Hardware-level investigation: verify heater operating point, characterize surface morphology; possibly isothermal calibration
ML-level remediability	High	Low; physics-level investigation required

12.2 The Broader Scientific Contribution

The central scientific contribution of this project is the demonstration that calibration transfer performance is governed not only by the magnitude of domain shift but by the underlying failure mechanism — and that different mechanisms require different remedies.

Previous calibration transfer studies typically measure transfer performance (RMSE, R^2) and report whether specific adaptation strategies improve it. This project goes one step further: it asks why strategies succeed or fail, and constructs a diagnostic framework that links the pattern of failure to its physical cause and its appropriate remedy.

The result is not merely academic. In a deployment scenario where a new sensor board must be characterized before calibration strategy selection, the diagnostic score framework from Day 5 provides a quantitative basis for this decision. A high coverage score and quick adaptation response identifies a board as B5-type: a few labeled calibration points and mean/std alignment will suffice. A low coverage score with persistent signed error at high concentration identifies a board as B1-type: software-level adaptation will not solve the problem, and hardware characterization is required.

12.3 The Null Results As Scientific Contributions

Three documented negative results from this project carry as much scientific weight as the positive findings:

- Geometry metrics do not predict transfer success. The intuitive assumption that boards with smaller PCA distance should transfer more easily was not supported. Coverage metrics are stronger predictors.
- Complex adaptation methods do not outperform simple alignment at realistic calibration budgets. Physics-aware piecewise corrections, CORAL alignment, and quadratic residual models all failed to consistently improve over mean/std alignment with 1–10 target samples.
- B1 is not fully rescuable through any tested adaptation strategy. The recognition that some transfer failures require hardware investigation, not software adaptation, is a practically important negative result.

Each of these null results changes the engineering decision space. They reduce investment in approaches that will not work and clarify which problems require solutions at a different level of the system.

13. Practical Deployment Implications

13.1 Recommended Calibration Protocol

Based on the full experimental record, a practical calibration transfer protocol for MOx gas sensor arrays in combustible gas detection applications is:

- Use physics-informed features as the primary representation. Normalized Rs/R_0 ratios and response-magnitude descriptors outperform raw resistance values in zero-shot transfer. Implement these as a standard pre-processing stage.
- Identify board-invariant feature subsets (from Day 5 transferability analysis) and use these as the core feature set for field deployment. Features with low board-to-board coefficient of variation and monotonic concentration response reduce recalibration burden.
- Apply the Day 5 diagnostic framework to any new target board before selecting an adaptation strategy. Compute coverage score and evaluate signed prediction error pattern. Classify the board as coverage-limited or potentially target-intrinsic.

- For coverage-limited boards (B5-type): collect 1–2 labeled calibration measurements at well-distributed concentrations. Apply mean/std feature alignment. This will resolve >80% of transfer error.
- For target-intrinsic boards (B1-type): recognize that software-level adaptation has limited effectiveness. Escalate to hardware characterization: verify heater operating point, measure impedance spectroscopy profile, compare surface morphology data if available.
- Treat low-concentration calibration carefully. Day 4 showed that low-concentration transfer can be as difficult as high-concentration transfer due to lower SNR and stronger baseline drift effects.

13.2 Calibration Efficiency

The finding that mean/std alignment achieves >80% RMSE reduction on coverage-limited boards with a single labeled sample has direct implications for calibration cost. A full factory recalibration of a sensor board — requiring multi-point gas exposures across the full concentration range — can be replaced by a single reference-point exposure at one or two concentrations, with the remaining calibration performed in software using the source-board model and mean/std alignment.

This is particularly relevant for field-deployed systems where recalibration infrastructure is limited. A combustible gas detector in a remote or hazardous environment can be recalibrated with a single reference gas exposure, dramatically reducing maintenance cost and downtime.

13.3 Limitations and Future Directions

Several important limitations bound the conclusions of this study:

- Dataset scope: The UCI Twin Gas Sensor Arrays dataset contains five boards with limited replicate measurements per board, particularly for B5. The 0-shot vs. 1-shot conclusions are robust; multi-shot scaling analysis (2-shot, 5-shot, 10-shot) is underpowered.
- Single gas focus: The analysis concentrated on methane. Calibration transfer behavior may differ for the other gases in the dataset (CO, ethanol, ethylene), and multi-gas discrimination adds a further layer of complexity.
- Physical mechanism of B1: The surface-site saturation, heater variation, and microstructural hypotheses advanced in Section 7 are consistent with the observations but have not been experimentally confirmed. Controlled characterization experiments would be needed.
- Generalizability: The failure-mode taxonomy was derived from one dataset. Validation on additional datasets — particularly with sensor boards spanning a wider range of manufacturing conditions — would strengthen the generality of the diagnostic framework.

Future directions include: investigation of monotonic calibration mappings for B1-type boards (potentially correcting response compression while preserving concentration rank-ordering), evaluation of anchor-point calibration protocols (which concentration points provide the most information per calibration measurement), and extension of the diagnostic framework to multi-gas scenarios.

14. Conclusions

This project investigated calibration transfer for MOx combustible gas detection through eight systematic experimental phases, from zero-shot baseline characterization through few-shot adaptation to formal failure-mode analysis. The central finding is:

> ***Calibration transfer performance is governed not only by domain-shift magnitude, but by the underlying failure mechanism. Two boards can exhibit similar transfer error while requiring fundamentally different remedies. This distinction is the project's core scientific contribution.***

The specific conclusions are:

- Physics-informed feature engineering improves zero-shot transfer performance. Normalized Rs/R_0 features and response-magnitude descriptors achieve $R^2 = 0.957$ on a held-out board versus $R^2 = 0.921$ for raw features — a difference attributable entirely to feature design, with no target calibration data required.
- Coverage predicts transfer success better than geometric proximity. Source-domain coverage metrics (explained variance volume, effective dimensionality, high-concentration coverage) are stronger predictors of transfer RMSE than PCA centroid distance or covariance distance. Source-board diversity should be optimized for coverage, not similarity.
- Board B5 represents a coverage-limited failure mode. Its transfer error collapses by more than 80–90% with a single labeled target calibration point, consistent with mean/std feature alignment correcting the dominant low-order statistical mismatch between source and target feature distributions.
- Board B1 represents a target-intrinsic failure mode. Its systematic concentration-dependent response compression persists through all tested adaptation strategies. The physical mechanism — most likely surface-site saturation, heater operating-point variation, or microstructural differences — cannot be corrected by software adaptation alone.
- Simple statistical alignment outperforms complex physics-aware corrections at realistic calibration budgets. At 1–10 target calibration samples, global mean/std alignment dominates piecewise corrections, CORAL alignment, and quadratic residual models. The dominant board-to-board mismatch is low-order statistical deformation, not structured nonlinear distortion.
- Label leakage, when present, produces qualitatively incorrect scientific conclusions. The corrected analysis reversed a key Day 4 conclusion and underscores the importance of feature-set auditing in ML-based sensor calibration development.

The practical recommendation is a single, deployable decision rule:

> ***Classify the failure mode before choosing a remediation strategy. Coverage-limited boards respond to few-shot statistical alignment. Intrinsically compressed boards require hardware-level investigation and characterization.***

This recommendation does not require sophisticated machine learning or complex hardware. It requires understanding the sensor physics, applying appropriate feature engineering, collecting a small number of diagnostic calibration measurements, and interpreting the results through the failure-mode framework developed in this project.

A model trained on some boards can transfer reliably to others — provided we understand why and when it fails. That understanding is the engineering contribution of this work.

References and Methods Summary

Dataset Citation

Fonollosa, J., Fernández, L., Gutiérrez-Gálvez, A., Huerta, R., & Marco, S. (2015). Calibration transfer and drift counteraction in chemical sensor arrays using Direct Standardization. Sensors and Actuators B: Chemical, 236, 1044–1053. https://doi.org/10.1016/j.snb.2016.05.089

Dataset: UCI Machine Learning Repository, Twin Gas Sensor Arrays, ID #361. Available under CC BY 4.0.

Methods Summary

Category	Methods Used
Regression	Ridge Regression, RandomForest, XGBoost
Feature engineering	Statistical descriptors (min, max, std, percentiles); physics-informed (Rs/R_0, normalized response, drift-corrected baseline); combined sets
Domain adaptation	Mean/std feature alignment, linear recalibration, residual correction, CORAL covariance alignment (negative result), piecewise recalibration (negative result)
Few-shot adaptation	0-shot through 10-shot calibration using labeled target-board samples
Manifold analysis	PCA (board trajectory, manifold overlap, coverage volume, centroid distance, covariance distance)
Failure mode diagnostics	Coverage score, gain mismatch score, curvature mismatch score, intrinsic deformation score, high-concentration compression score
Tools / Libraries	Python 3.10, scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, SciPy
Excluded approaches	Deep learning, UMAP/t-SNE, large hyperparameter searches — intentionally excluded for deployment realism

— End of Report —